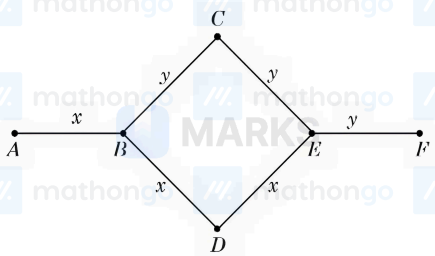
Q1. JEE Main 2026 (22 January Shift 1)

Rods x and y of equal dimensions but of different materials are joined as shown in figure. Temperatures of end points A and F are maintained at 100°C and 40°C respectively. Given the thermal conductivity of rod x is three times of that of rod y, the temperature at junction points B and E are (close to):



(1) 80°C and 70°C respectively
(2) 80°C and 60°C respectively
(3) 60°C and 45°C respectively
(4) 89°C and 73°C respectively

Q2. JEE Main 2026 (21 January Shift 1)

An aluminium and steel rods having same lengths and cross-sections are joined to make total length of 120 cm at 30°C. The coefficient of linear expansion of aluminium and steel are $24 \times 10^{-6}/°C$ and $1.2 \times 10^{-5}/°C$, respectively. The length of this composite rod when its temperature is raised to 100°C, is ____ cm.

(1) 120.20 (2) 120.15 (3) 120.06 (4) 120.03

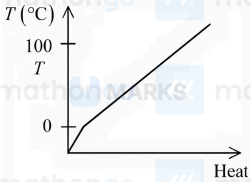
Q3. JEE Main 2026 (21 January Shift 1)

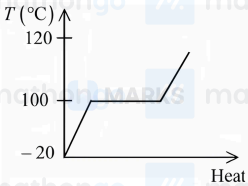
A gas based geyser heats water flowing at the rate of 5.0 litres per minute from 27°C to 87°C. The rate of consumption of the gas is ____ g/s.
(Take heat of combustion of gas $= 5.0 \times 10^4$ J/g) specific heat capacity of water $= 4200$ J/kg · °C

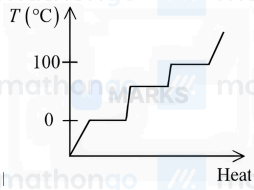
(1) 0.42 (2) 2.1 (3) 0.21 (4) 4.2

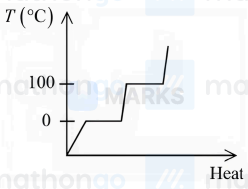
Q4. JEE Main 2026 (28 January Shift 1)

Which of the following best represents the temperature versus heat supplied graph for water, in the range of −20°C to 120°C ?

(1)


(2)


(3)


(4)


Q5. JEE Main 2026 (28 January Shift 1)

10 kg of ice at $-10°C$ is added to 100 kg of water to lower its temperature from 25 °C. Consider no heat exchange to surroundings. The decrement to the temperature of water is ____ °C.
(specific heat of ice $= 2100\ J/Kg.°C$, specific heat of water $= 4200\ J/Kg.°C$, latent heat of fusion of ice $= 3.36 \times 10^5\ J/Kg$)

(1) 6.67 (2) 11.6 (3) 15 (4) 10

Q6. JEE Main 2026 (24 January Shift 1)

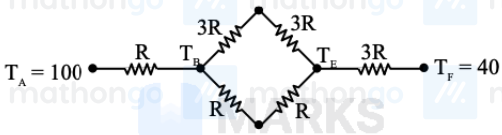
A brass wire of length 2 m and radius 1 mm at $27°C$ is held taut between two rigid supports. Initially it was cooled to a temperature of $-43°C$ creating a tension T in the wire. The temperature to which the wire has to be cooled in order to increase the tension in it to $1.4T$, is ____ °C.

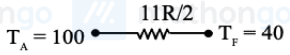
(1) -86 (2) -65 (3) -71 (4) -80

ANSWERS AND SOLUTIONS

1. (4) **2.** (2) **3.** (1) **4.** (4) **5.** (4) **6.** (3)

1. **(4)** Let $\left[R = \frac{\ell}{3KA}\right]$





$$\left[H = \frac{100-40}{\frac{11R}{2}}\right] \text{...(1)}$$

$$H = \frac{100 - T_B}{R} \text{.....(2)}$$

$$H = \frac{T_E - 40}{3R} \text{.....(3)}$$

using (1) and (2)

$120 = 1100 - 11T_A$

$T_B = 89^\circ C$

using (1) and (3)

$T_E = 73^\circ C$

2. **(2)** Total length at 30°C = 120 cm, each rod = 60 cm

$\alpha_{Al} = 24 \times 10^{-6}$/°C, $\alpha_{Steel} = 1.2 \times 10^{-5}$/°C

$\Delta T = 100 - 30 = 70$°C

$\Delta L_{Al} = 60 \times 24 \times 10^{-6} \times 70 = 0.1008$ cm

$\Delta L_{Steel} = 60 \times 1.2 \times 10^{-5} \times 70 = 0.0504$ cm

Total expansion $= 0.1008 + 0.0504 = 0.1512$ cm

Final length $= 120 + 0.1512 = 120.15$ cm

3. **(1)** Flow rate = 5.0 L/min = 5.0 kg/min, $\Delta T = 87 - 27 = 60$°C

Heat required per minute: $Q = mc\Delta T = 5.0 \times 4200 \times 60 = 1.26 \times 10^6$ J/min

If m grams of gas consumed per minute:

$m \times 5.0 \times 10^4 = 1.26 \times 10^6$

$m = \frac{1.26 \times 10^6}{5.0 \times 10^4} = 25.2$ g/min

Rate $= \frac{25.2}{60} = 0.42$ g/s

4. **(4)** No solution.

5. **(4)** Heat available from water cooling 25°C to 0°C: $Q_{available} = 100 \times 4200 \times 25 = 10.5 \times 10^6$ J.
Heat to warm ice from -10°C to 0°C: $Q_1 = 10 \times 2100 \times 10 = 0.21 \times 10^6$ J.
Heat to melt ice at 0°C: $Q_2 = 10 \times 3.36 \times 10^5 = 3.36 \times 10^6$ J.
Total needed: $0.21 + 3.36 = 3.57 \times 10^6$ J.
Since available heat exceeds required heat, all ice melts. Remaining heat: $10.5 - 3.57 = 6.93 \times 10^6$ J
warms 110 kg water: $\Delta T = \frac{6.93 \times 10^6}{110 \times 4200} = 15$ °C.
Final temperature: 15°C.
Decrement in water temperature: $25 - 15 = 10$ °C.

6. **(3)** Thermal stress in the wire is caused by contraction.
At -43°C, the temperature drop from 27°C is $\Delta T_1 = 70^\circ C$
It creates tension $T = YA\alpha\Delta T_1$ where Y is Young's modulus and α is linear expansion coefficient.
For tension to increase to $1.4T$, the new temperature drop must be: $\Delta T_2 = 1.4 \times 70 = 98^\circ C$.
Required temperature: $T_3 = 27 - 98 = -71^\circ C$